

DELTA RPI-3 PHASE INVERTER

MODBUS Definition



General Information	
Baud Rate	9600/19200(Default)/38400bps
Parity	None
Data Bits	8
Stop Bits	1
Addresses Mode	PLC Addresses Mode (Base 1)

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1. Message Format

Address	Function Code	Data	Error Check
1 byte	1 bytes	<=252 byte	2 bytes

1.1 Address

PCS ID=1~255 (Default : 1)(255 is for broadcast only)

1.2 Function Code

0x03: Read Holding Register (16-bit word , R/W)

0x04: Read Input Register (16-bit word , R)

0x05: Write Single Coil

0x06: Write Single Register

0x10: Write Multiple Registers

1.3 Data

1.4 Error Check

CRC

2. MODBUS Message Format

2.1 Read Holding Registers (0x03)

INQUIRY : Computer → PCS

Address	Function Code	Data				Error Check	
1~255	0x03	Startin g Addres s HI	Startin g Addres s LO	No of Register s HI	No of Register s LO	Error Check LO	Error Check HI

RESPONSE : PCS → Computer

Address	Function Code	Data						Error Check	
1~255	0x03	Byte Count	Dat a HI	Dat a LO	Dat a HI	Dat a LO		Error Check LO	Error Check HI

ERROR: PCS → Computer

Address	Function Code	Data	Error Check	
1~255	0x83	0x01 or 0x02 or 0x03 or 0x04	Error Check LO	Error Check HI

PS. starting Address : 0x0000 to 0xFFFF

No of Registers : 1 to 125 (0x7D)

2.2 Read Input Registers (0x04)

INQUIRY : Computer → PCS

Address	Function Code	Data				Error Check	
1~255	0x04	Startin g Addres s HI	Startin g Addres s LO	No of Register s HI	No of Register s LO	Error Check LO	Error Check HI

RESPONSE : PCS → Computer

Address	Function Code	Data						Error Check	
1~255	0x04	Byte Count	Dat a HI	Dat a LO	Dat a HI	Dat a LO		Error Check LO	Error Check HI

ERROR: PCS → Computer

Address	Function Code	Data	Error Check	
1~255	0x84	0x01 or 0x02 or 0x03 or 0x04	Error Check LO	Error Check HI

PS. starting Address : 0x0000 to 0xFFFF

No of Registers : 1 to 125 (0x7D)

2.3 Write Single Coil (0x05)

INQUIRY : Computer → PCS

Address	Function Code	Data				Error Check	
1~255	0x05	Coil Address HI	Coil Address LO	Write Data HI	Write Data LO	Error Check LO	Error Check HI

RESPONSE : PCS → Computer

Address	Function Code	Data				Error Check	
1~255	0x05	Coil Address HI	Coil Address LO	Write Data HI	Write Data LO	Error Check LO	Error Check HI

ERROR: PCS → Computer

Address	Function Code	Data				Error Check	
1~255	0x85	0x01 or 0x02 or 0x03 or 0x04				Error Check LO	Error Check HI

PS. Coil Address : 0x0000 to 0xFFFF

Write Data : 0x0000 or 0xFF00

2.4 Write Single Register (0x06)

INQUIRY : Computer → PCS

Address	Function Code	Data				Error Check	
1~255	0x06	Register Address HI	Register Address LO	Write Data HI	Write Data LO	Error Check LO	Error Check HI

RESPONSE : PCS → computer

Address	Function Code	Data				Error Check	
1~255	0x06	Register Address HI	Register Address LO	Write Data HI	Write Data LO	Error Check LO	Error Check HI

ERROR: PCS → computer

Address	Function Code	Data				Error Check	
1~255	0x86	0x01 or 0x02 or 0x03 or 0x04				Error Check LO	Error Check HI

PS. Register Address : 0x0000 to 0xFFFF

Write Data : 0x0000 or 0xFFFF

2.5 Write Multiple Registers (0x10)

INQUIRY : Computer → PCS

Address	Function Code	Data							Error Check	
1~255	0x10	Starting Address HI	Starting Address LO	No of Registers HI	No of Registers LO	Byte Count	Registers Value		Error Check LO	Error Check HI

RESPONSE : PCS → Computer

Address	Function Code	Data				Error Check	
1~255	0x10	Starting Address HI	Starting Address LO	No of Registers HI	No of Registers LO	Error Check LO	Error Check HI

ERROR : PCS → Computer

Address	Function Code	Data	Error Check	
1~255	0x90	0x01 or 0x02 or 0x03 or 0x04	Error Check LO	Error Check HI

PS. starting Address : 0x0000 to 0xFFFF

No of Registers : 0x0001 to 0x0078

3. Memory Map ~ Read Only (Input Registers)

<SYSTEM INFORMATION>

New Address (Hex)	New Address (Dec)	Description	
408	1032	<i>Inverter ID</i>	<i>RS485 ID</i>
409	1033	<i>RTC Year</i>	
40A	1034	<i>RTC Month</i>	
40B	1035	<i>RTC Day</i>	
40C	1036	<i>RTC Hour</i>	
40D	1037	<i>RTC Minute</i>	
40E	1038	<i>RTC Second</i>	
40F	1039	<i>Reserved</i>	
A003	40963	<i>FW1 Revision</i>	<i>ex : 0x0A0B => V10.11</i>
A004	40964	<i>FW1 Date</i>	<i>ex : 0x082D => Y08W45</i>
A005	40965	<i>FW2 Revision</i>	<i>ex : 0x0A0B => V10.11</i>
A006	40966	<i>FW2 Date</i>	<i>ex : 0x082D => Y08W45</i>
A007	40967	<i>FW3 Revision</i>	<i>ex : 0x0A0B => V10.11</i>
A008	40968	<i>FW3 Date</i>	<i>ex : 0x082D => Y08W45</i>
416	1046	<i>Reserved</i>	
417	1047	<i>Reserved</i>	
D008	53256	<i>PCS Status</i>	<i>REFER TO APPENDIX C</i>
D00A	53258	<i>Reconnected Time</i>	<i>unit : second</i>
41A	1050	<i>Device Type</i>	<i>unit : 100W ; ex : 200 => 20KW, 150 => 15KW, 0 => default(searching)</i>

<AC INFORMATION>

Address (Hex)	Address (Dec)	Purpose	Description
C000	49152	<i>Total Power</i>	<i>Unit=1W</i>
C001	49153	<i>AC Voltage - 1</i>	<i>Unit=0.1V</i>
C002	49154	<i>AC Current - 1</i>	<i>Unit=0.01A</i>
C003	49155	<i>AC Power - 1</i>	<i>Unit=1W</i>
C004	49156	<i>AC Frequency - 1</i>	<i>Unit=0.01Hz</i>
C005	49157	<i>AC Voltage - 2</i>	<i>Unit=0.1V</i>
C006	49158	<i>AC Current - 2</i>	<i>Unit=0.01A</i>
C007	49159	<i>AC Power - 2</i>	<i>Unit=1W</i>
C008	49160	<i>AC Frequency - 2</i>	<i>Unit=0.01Hz</i>
C009	49161	<i>AC Voltage - 3</i>	<i>Unit=0.1V</i>
C00A	49162	<i>AC Current - 3</i>	<i>Unit=0.01A</i>
C00B	49163	<i>AC Power - 3</i>	<i>Unit=1W</i>
C00C	49164	<i>AC Frequency - 3</i>	<i>Unit=0.01Hz</i>

<DC INFORMATION>

New Address (Hex)	New Address (Dec)	Purpose	Description
B000	45056	<i>Total Power</i>	<i>Unit=1W</i>
B001	45057	<i>Input (Solar) Voltage-1</i>	<i>Unit=0.1V</i>
B002	45058	<i>Input (Solar) Current-1</i>	<i>Unit=0.01A</i>
B003	45059	<i>Input (Solar) Power-1</i>	<i>Unit=1W</i>
B004	45060	<i>Input (Solar) Voltage-2</i>	<i>Unit=0.1V</i>
B005	45061	<i>Input (Solar) Current-2</i>	<i>Unit=0.01A</i>
B006	45062	<i>Input (Solar) Power-2</i>	<i>Unit=1W</i>

<Harvest Energy / Run Time / Inside Temperature>

New Address (Hex)	New Address (Dec)	Description	
D000	53248	Today Energy	1072:Low Word, 1073: High Word unit : 10Wh
D001	53249		
D002	53250	Today Runtime	1074:Low Word, 1075: High Word unit : 1 second
D003	53251		
D004	53252	Life Energy	1076:Low Word, 1077: High Word unit : 10Wh
D005	53253		
D006	53254	Life Runtime	1078:Low Word, 1079: High Word unit : 1 second
D007	53255		
D008	53259	Temperature 1	unit : 1C
D009	53260	Temperature 2	unit : 1C
D00A	53261	Temperature 3	unit : 1C
D00B	53262	Temperature 4	unit : 1C

<EVENT LIST>

Address (Hex)	Address (Dec)	Description	
440	1088	Event Index	This shall be assigned via Holding Registers (801) first when reading the events. The event index is the page index.
441	1089	Event (index*10+0)	
442	1090	Event (index*10+1)	
443	1091	Event (index*10+2)	For current design in PV inverter the maximum event is 30, so the index is from 0 to 2. If the index is equal to 0, the event with index from 0 to 9 will be reported. If the index is equal to 1, the event with index from 10 to 19 will be reported, and so on.
444	1092	Event (index*10+3)	
445	1093	Event (index*10+4)	
446	1094	Event (index*10+5)	
447	1095	Event (index*10+6)	
448	1096	Event (index*10+7)	
449	1097	Event (index*10+8)	
44A	1098	Event (index*10+9)	

<SERIAL NUMBER>

Address (Hex)	Address (Dec)	Description
470	1136	<i>Serial number 0,1</i>
471	1137	<i>Serial number 2,3</i>
472	1138	<i>Serial number 4,5</i>
473	1139	<i>Serial number 6,7</i>
474	1140	<i>Serial number 8,9</i>
475	1141	<i>Serial number 10,11</i>
476	1142	<i>Serial number 12,13</i>
477	1143	<i>Serial number 14,15</i>

<CURRENT EVENT>

Address (Hex)	Address (Dec)	Purpose	Description	
C00	3072	Current even - Error0	E15~E00	ex : E15 E14 E13E01 E00 0100 1000 0000 0010
C01	3073	Current even - Error1	E31~E16	E31 E14 E13E17 E16
C02	3074	Current even - Error2	E47~E32	
C03	3075	Current even - Error3		
C04	3076	Current even - Error4		
C05	3077	Current even - Error5		
C06	3078	Current even - Error6		
C07	3079	Current even - Error7		
C08	3080	Current even - Error8		
C09	3081	Current even - Error9		
C0A	3082	Current even - Error10		
C0B	3083	Current even - Error11		
C0C	3084	Current even - Error12		
C0D	3085	Current even - Error13		
C0E	3086	Current even - Error14		
C0F	3087	Current even - Error15		
C10	3088	Current even - Warning0	W15~W00	
C11	3089	Current even - Warning1	W31~W16	
C12	3090	Current even - Warning2	W47~W32	
C13	3091	Current even - Warning3		
C20	3104	Current even - Fault0	F15~F00	
C21	3105	Current even - Fault1	F31~F16	
C22	3106	Current even - Fault2	F47~F32	
C23	3107	Current even - Fault3		
C24	3108	Current even - Fault4		
C25	3109	Current even - Fault5		
C26	3110	Current even - Fault6		
C27	3111	Current even - Fault7		
C28	3112	Current even - Fault8		
C29	3113	Current even - Fault9		
C2A	3114	Current even - Fault10		
C2B	3115	Current even - Fault11		
C2C	3116	Current even - Fault12		
C2D	3117	Current even - Fault13		
C2E	3118	Current even - Fault14		
C2F	3119	Current even - Fault15		

※PLEASE REFER TO APPENDIX D

<DAILY ENERGY LOG>

Address (Hex)	Address (Dec)	Purpose	Description
800	2048	Day-0 Wh (Today)	2048: Low word,2049:High Word unit : Wh
801	2049		
802	2050	Day-1 Wh (Yesterday)	unit : Wh
803	2051		
804	2052	Day-2 Wh	unit : Wh
805	2053		
806	2054	Day-3 Wh	unit : Wh
807	2055		
808	2056	Day-4 Wh	unit : Wh
809	2057		
80A	2058	Day-5 Wh	unit : Wh
80B	2059		
80C	2060	Day-6 Wh	unit : Wh
80D	2061		
80E	2062	Day-7 Wh	unit : Wh
80F	2063		
810	2064	Day-8 Wh	unit : Wh
811	2065		
812	2066	Day-9 Wh	unit : Wh
813	2067		
814	2068	Day-10 Wh	unit : Wh
815	2069		
816	2070	Day-11 Wh	unit : Wh
817	2071		
818	2072	Day-12 Wh	unit : Wh
819	2073		
81A	2074	Day-13 Wh	unit : Wh
81B	2075		
81C	2076	Day-14 Wh	unit : Wh
81D	2077		
81E	2078	Day-15 Wh	unit : Wh
81F	2079		
820	2080	Day-16 Wh	unit : Wh
821	2081		
822	2082	Day-17 Wh	unit : Wh
823	2083		
824	2084	Day-18 Wh	unit : Wh
825	2085		
826	2086	Day-19 Wh	unit : Wh
827	2087		
828	2088	Day-20 Wh	unit : Wh
829	2089		
82A	2090	Day-21 Wh	unit : Wh
82B	2091		
82C	2092	Day-22 Wh	unit : Wh
82D	2093		

82E	2094	Day-23 Wh	unit : Wh
82F	2095		
830	2096	Day-24 Wh	unit : Wh
831	2097		
832	2098	Day-25 Wh	unit : Wh
833	2099		
834	2100	Day-26 Wh	unit : Wh
835	2101		
836	2102	Day-27 Wh	unit : Wh
837	2103		
838	2104	Day-28 Wh	unit : Wh
839	2105		
83A	2106	Day-29 Wh	unit : Wh
83B	2107		
83C	2108	Day-30 Wh	unit : Wh
83D	2109		
83E	2110	Day-31Wh	unit : Wh
83F	2111		

4. Memory Map ~ Writable (Holding Register)

<Event Index>

Address (Hex)	Address (Dec)	Description
321	801	<i>Event Index</i>

<Manual Reset>

Address (Hex)	Address (Dec)	Description
6343	25411	<i>This value is set to "1" when the AC event happened if the PCS is in Manual Reset mode. Clear the event by filling into the "1".</i>

<DNO Control>

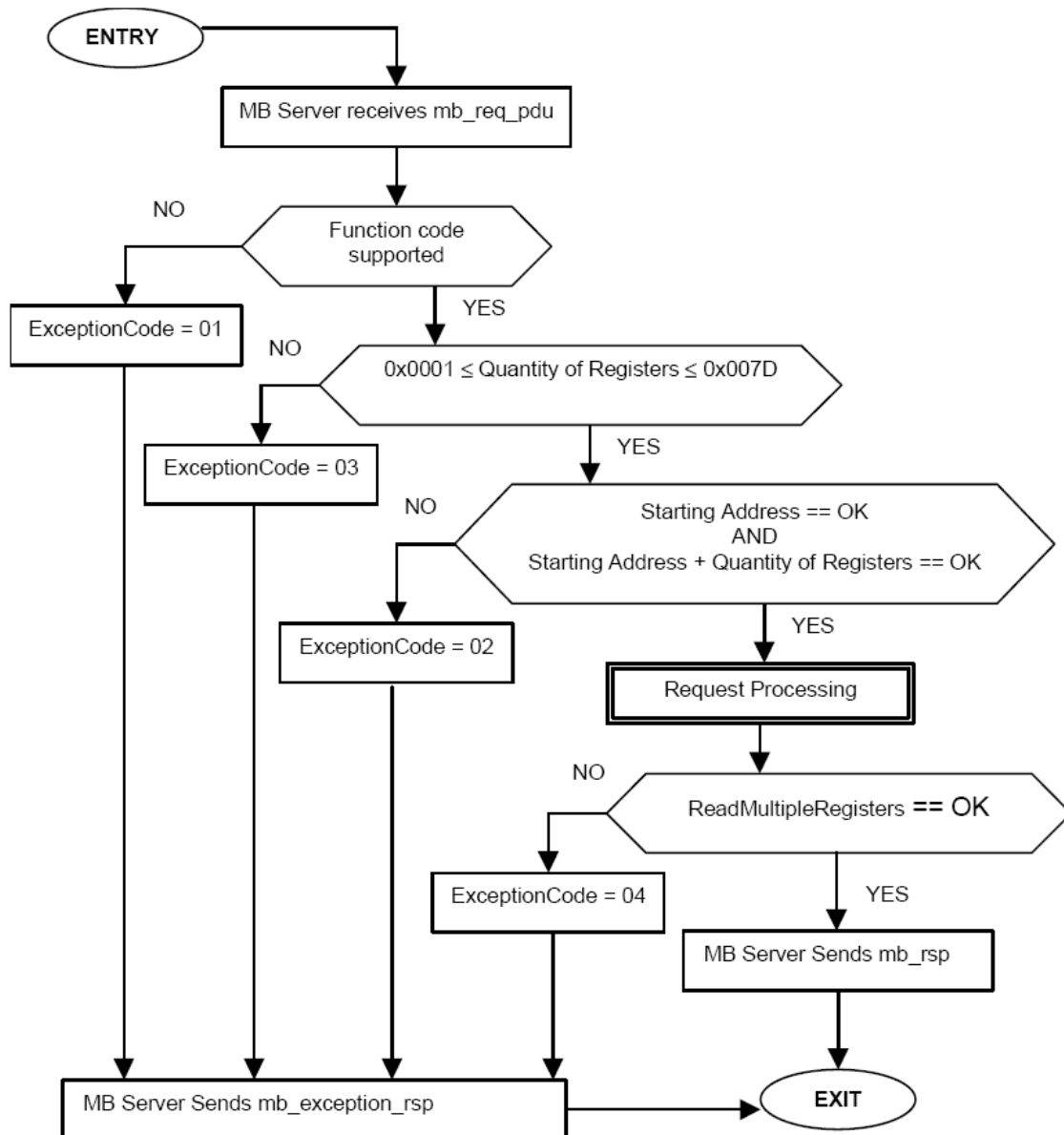
Address (Hex)	Address (Dec)	Description	
BD80	48512	PM_Mode	0: Disable ; 1: Enable
BD81	48513	PM_Percent	Range 0 ~ 100 ; unit 1%
BD82	48514	Active Power Slop	Range 60 ~ 600 ; unit 1sec

APPENDIX A : MODBUS Exception Responses

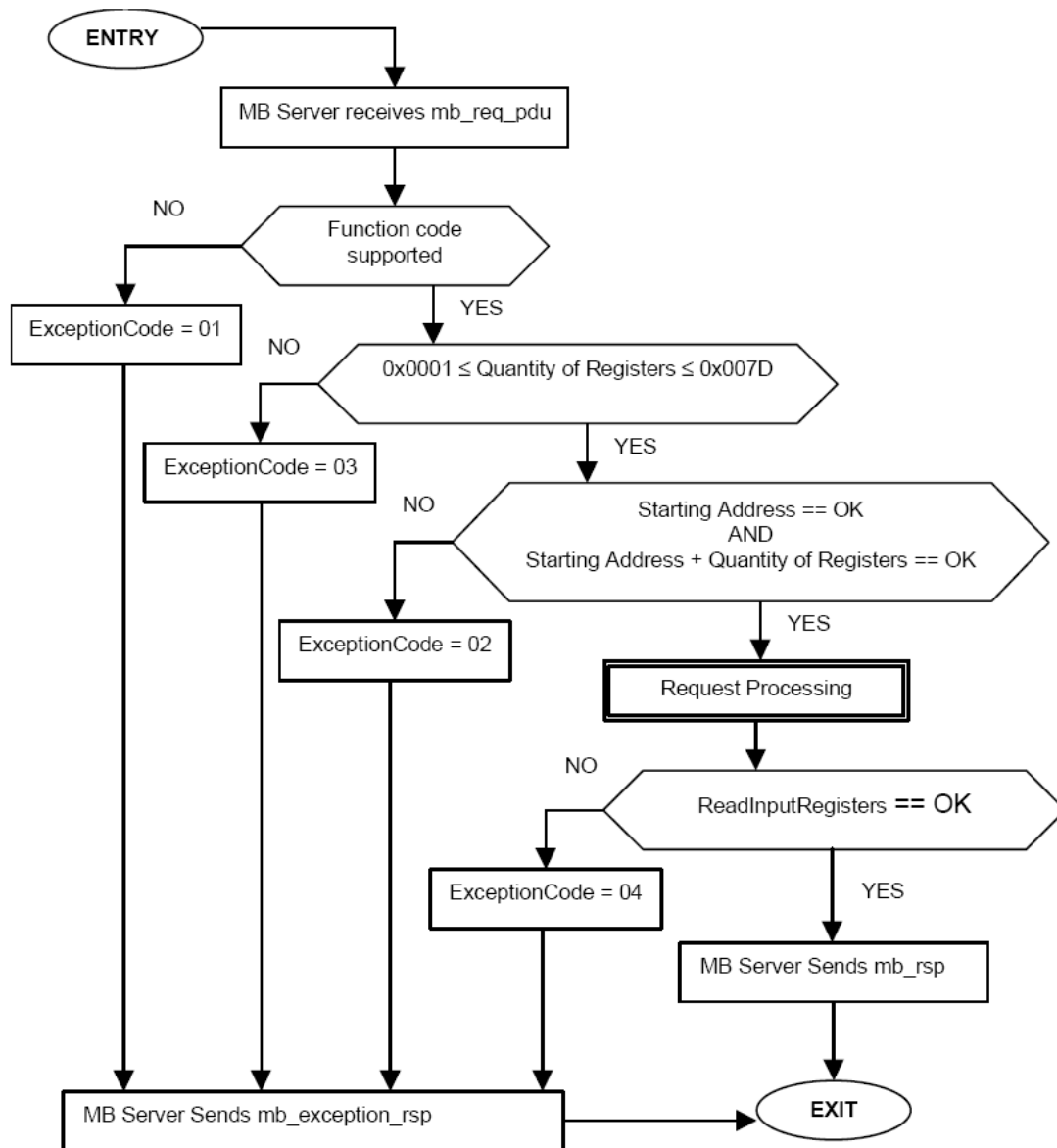
MODBUS Exception Codes		
Code	Name	Meaning
01	ILLEGAL FUNCTION	The function code received in the query is not an allowable action for the server (or slave).
02	ILLEGAL DATA ADDRESS	The data address received in the query is not an allowable address for the server (or slave).
03	ILLEGAL DATA VALUE	A value contained in the query data field is not an allowable value for server (or slave).
04	SLAVE DEVICE FAILURE	An unrecoverable error occurred while the server (or slave) was attempting to perform the requested action.

APPENDIX B : Communications Transmission Process

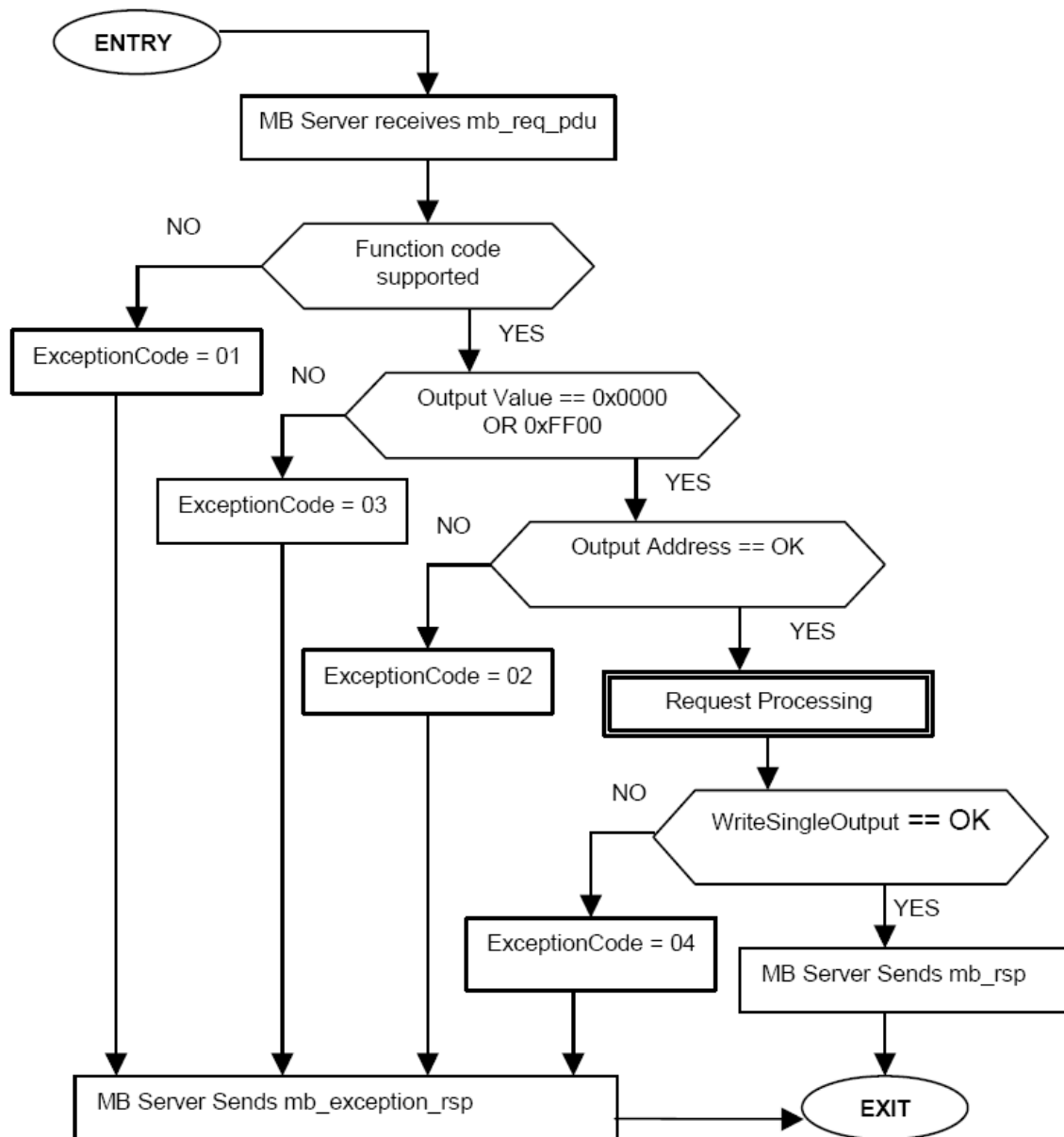
i. Function 0x03 Read Holding Registers



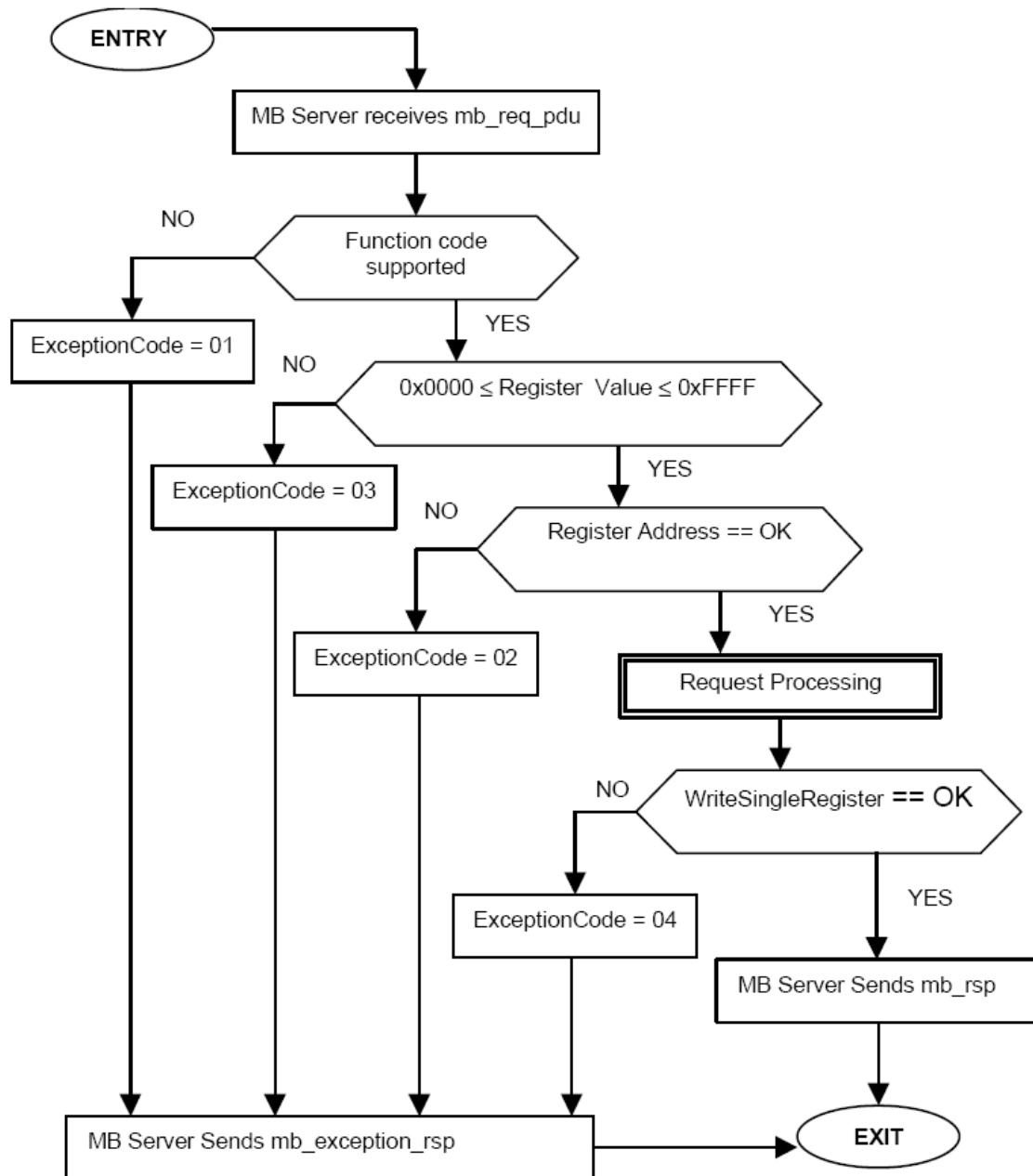
ii. Function 0x04 Read Input Registers



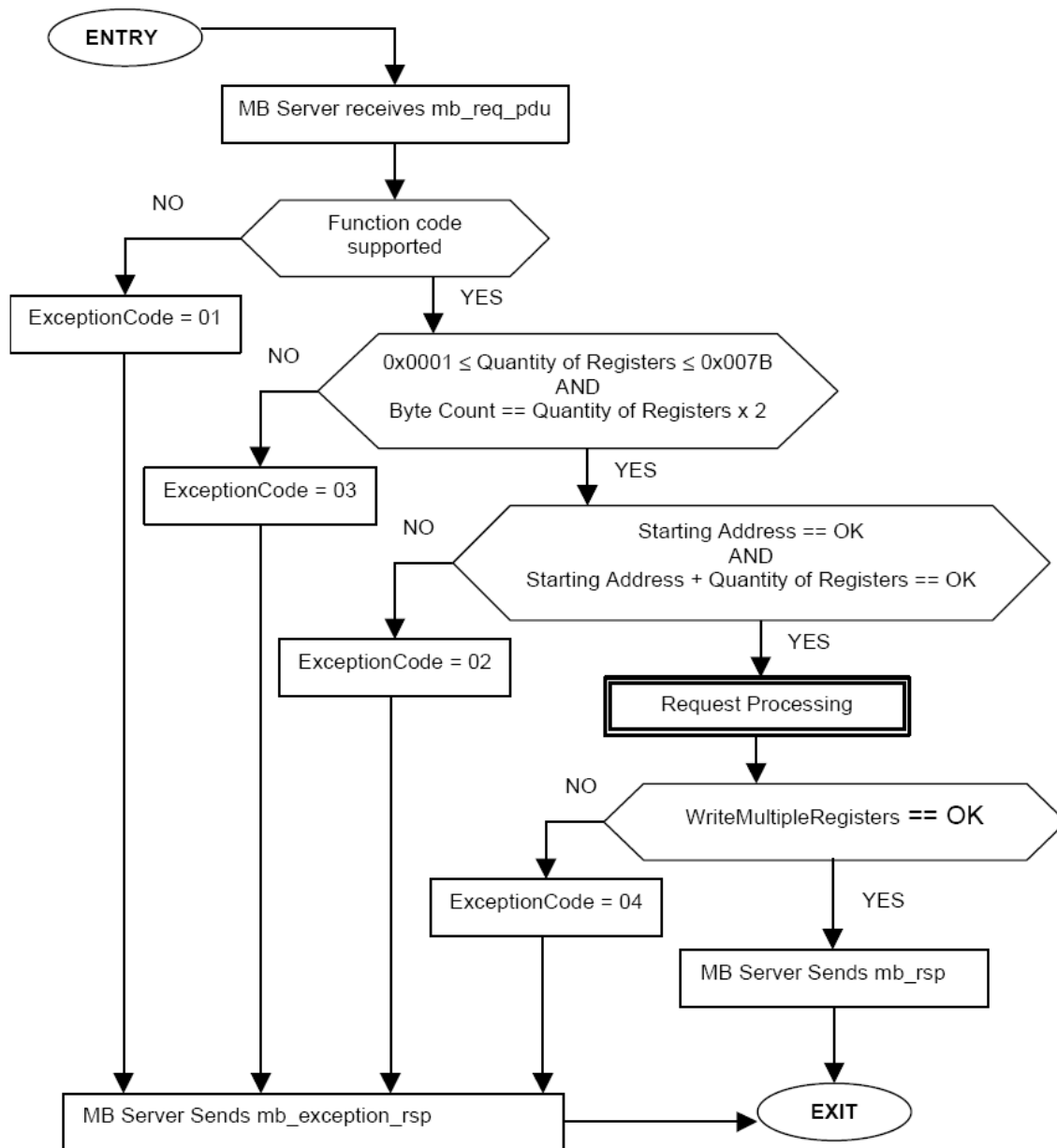
iii. Function 0x05 Write Single Coil



iv. Function 0x06 Write Single Register



v. Function 0x10 Write Multiple registers



APPENDIX C : PCS STATUS

PCS STATUS	Condition
0	Grid Tie - Standby
1	Grid Tie - System count down
2	Grid Tie - Operation
3	Grid Tie - No DC
4	Grid Tie - Alarm

APPENDIX D : EVENT CODE TABLE

Item	Value	Description
F01	1	DC Injection (R)
F02	2	DC Injection (S)
F03	3	DC Injection (T)
F04	4	DC Injection Total
F05	5	NTC OTP
F06	6	NTC0 Circuit Fail
F07	7	NTC LTP
F08	8	Ntc1 Circuit Fail
F09	9	Ntc2 Circuit Fail
F10	10	Ntc3 Circuit Fail
F13	13	Relay Open
F15	15	DSP ADC Vgrid/Iout Fail
F16	16	ADC Vin/Vbus Fail
F17	17	ADC Iin/Iboost Fail
F18	18	Red. ADC Vgrid Fail
F19	19	ADC Iout_dc Fail
F20	20	Efficiency Abnormal
F22	22	Internal Comm. Fault (Red.)
F23	23	Internal Comm. Fault (Dis.)
F24	24	RCMU Over Rating
F26	26	Three Phase Current Unbalance
F27	27	RCMU Circuit Fail
F28	28	Relay Test Short
F29	29	Relay Test Open
F30	30	Bus Unbalance
F31	31	Bus P Over Voltage
F33	33	Bus N Over Voltage
F35	35	Bus Voltage Over Rating
F36	36	OOCP-Transient (R)
F37	37	OOCP(R)
F38	38	OOCP-Transient (S)
F39	39	OOCP(S)
F40	40	OOCP-Transient (T)
F41	41	OOCP(T)
F42	42	CT sensor Fail (A)
F43	43	CT sensor Fail (B)
F44	44	CT sensor Fail (C)
F45	45	HW OOCF Circuit
F50	50	Zero Cross Circuit Fail
F60	60	IOCP(PV1)
F61	61	IOCP(PV2)
F70	70	IOCP-Transient(PV1)
F71	71	IOCP-Transient(PV2)

Item	Value	Description
E01	129	OFR
E02	130	UFR
E03	131	Islanding(Passive)
E04	132	Islanding(Active)
E06	134	Phase Jump
E07	135	Grid Quality
E08	136	Ac Connected Fail
E09	137	No Grid
E10	138	UVR(R)
E11	139	OVR(R)
E13	141	OVR-Slow(R)
E15	143	UVR(S)
E16	144	OVR(S)
E18	146	OVR-Slow(S)
E20	148	UVR(T)
E21	149	OVR(T)
E23	151	OVR-Slow(T)
E30	158	OVR(PV1)
E31	159	OVR(PV2)
E34	162	Insulation
W01	209	UVR (PV1)
W02	210	UVR (PV2)
W11	219	Fan Fail
W12	220	String Bus Comm.Fail
W13	221	Transformer Comm. Fail

----- END OF DOCUMENT -----